Intra-dimensional Working Memory Emission

Intake, source-grounded audit, and formal rehabilitation of IWME Draft 1.0

IWME-PRIME-01, Phases 0–3

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Phase 0–2 audit text from an earlier session (recorded as Fable 5)

Code and data: 9x25dillon/kgirl, package `src/kgirl/iwme`,
branch `claude/iwme-phase3`

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Abstract

IWME Draft 1.0 proposes that attention-mediated working memory can shape the evolution of a physical system and of the geometry an observer measures. This report preserves the draft, reconstructs its definitions and equations, audits them against the four supplied sources, applies two corrections the investigator made to the earlier audit, and rebuilds the core as a testable model. Every numerical statement here is computed by a 47-check registry in the `kgirl` repository; all checks pass. Five results carry the report. (i) The draft's four-equation system conflates observation with actuation and leaves its loop open; a repaired controlled state-space core separates them, and an observational-equivalence proposition shows that no statistic of the observables distinguishes relational memory from unmodeled physical state. (ii) In the attention–memory architecture, retention is set by the write rate alone for a fixed attention sequence; once the attention scores read the memory, the closed-loop Jacobian $(1-\rho)I + (\rho/T)\,\mathrm{Cov}_a(v,k)$ makes retention depend on the attention temperature T, with a pitchfork at $T_c = 1$, algebraic critical slowing $H_t \simeq (3/2\rho t)^{1/2}$, hysteresis of area $\propto (1-T)^2$, and flip or Neimark–Sacker crossings when the key–value geometry is anti-aligned or rotational. (iii) The Inverse Resonant Toroid of *The Geometry of Observation* is a parametrized 3-manifold in $\mathbb{R}^4$ whose Jacobian has rank three away from explicitly located circles; its published metric is wrong in three components, including a nonzero mixed term $g_{\varphi\psi}$, and its scalar curvature takes both signs, which refutes the published claim of negative curvature. (iv) The wake-field system of the Emergent Fractal Law source contains a false defect identity, a coherence equation that leaves $[0,1]$, a Lagrangian that generates a different field equation, and a first-order time scheme whose stability limit shrinks with damping. (v) An exact persistence-after-decoupling criterion is zero when memory cannot act on the physics, positive when it can, and separates an exponentially forgetting memory from an accumulator by its lag profile. Together they locate where evidence that memory acts on matter could come from; the report itself contains no such evidence.

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1 Scope, status, and how to reproduce

Object. IWME Draft 1.0 (Source A) is preserved verbatim as an archival object; nothing below rewrites it. This report has three layers: the Phase 0–2 audit (intake, reconstruction, adversarial audit), restated with the investigator's two corrections; audits of Sources B, C and D, which were unavailable when the Phase 0–2 report was written and have now been supplied; and Phase 3, the formal rehabilitation of the core architecture authorized by the investigator.

Investigator input applied.

(a) *Correction to Proposition P2.* The attention-independence of forgetting holds for the exponential moving average with inputs fixed in advance. When attention scores depend on memory, the system is a closed-loop nonlinear recurrence and attention changes the sensitivity of future memory to past memory; Proposition P3 already exhibits this. Applied as Propositions 6 and 7.

(b) *Correction to Finding F24.* The IRT has three parameters (θ, φ, ψ) embedded in four coordinates (x, y, z, w) with $w = -\int_0^{\psi} \kappa$. Such an embedding can have a Jacobian of rank three, and wherever it does, the induced three-dimensional metric is nondegenerate; what needs repair is the published metric, which must be recomputed with its nonzero mixed components. Applied in Section 5.1, which does the recomputation before anything else in this report uses the IRT.

(c) *Clarification of the hypothesis.* The temperature-controlled bifurcation of the illustrative attention model is an established type of bifurcation; the investigator adopts it as a rigorous mechanism to build on, without claiming a new physical law. Recorded in Section 6.2.

Evidence levels. Each claim carries a level: L0 established mathematics or engineering; L1 reasonable extrapolation; L2 hypothesis requiring experiment; L3 speculative. Every proposition proved here is L0. Every physical reading of IWME remains L2 or L3.

Reproduction. From the root of a `kgirl` checkout on branch `claude/iwme-phase3`:

- `PYTHONPATH=src python -m kgirl.iwme verify` runs the 47 checks in about 20 s (options `--fast`, `--json`, `--latex FILE`, `--only V22 V26`);
- `python -m unittest discover -s tests/iwme -t .` runs the 70 unit tests;
- `PYTHONPATH=src python -m kgirl.iwme plot -o research/iwme/figures` regenerates the five figures.

Check identifiers V01–V47 in the text point to Appendix A. All variates come from the repository's deterministic logistic-map generator (`kgirl.evolve.chaos.ChaosRNG`); no pseudo-random generator is used anywhere.

2 Phase 0: intake and provenance

Five documents enter this report (Table 1). Sources B–D are unpublished or non-peer-reviewed manuscripts and are cited as such. The attribution of Source C to the "Neurotronic Phase Caster

Research Collective" is taken from its title page.

Table 1: Sources. "Used by Draft 1.0" lists what the draft takes from each source; "Audited" says where this report checks it.

Source	Document	Used by Draft 1.0	Audited
A	IWME Foundational Draft 1.0	—	§§3–4
B	*The Emergent Fractal Law and the* 3^{627} *Framework* (anonymous, 21 Oct 2025, 8 pp.)	the wake equation (draft §6)	§5.2
C	*Geometry as Emergent Uni-telligence / The Geometry of Observation* (Neurotronic Phase Caster Research Collective, 7 Nov 2025, 19 pp.)	the history coordinate $w(\psi)$	§5.1
D	*The Convergence Point* (J. Edwards, with D. J. McConnell; 14 pp.)	named in the preamble, abstract and references	§5.3
E	*From Triangle to Form* (M. Oo et al., 2026; 25 pp.)	nothing	scope note, §5.4

3 Phase 1: reconstruction inventory

Definitions. D1, relational information state $H_t \in \mathcal{H}$ (no topology, dimension or probability structure given). D2, temporally relative environment E_t with local clock τ_E (the clock is never used again; E_t is said to contain physical state variables, which overlaps with X_t). D3, dimensional perspective $\Pi_{E,\lambda} : \mathcal{H} \to \mathcal{O}_{E,\lambda}$, an observation map. D4, working memory emission $G_{t+1} = T_{E_t,\lambda}(X_{t+1}, H_{t+1}, A_t)$. D5, operational presence (verbal; operationalized only as R in draft §7.2). "Intra-dimensional" and "emission" are never defined.

Equations (labels assigned here). E1 $X_{t+1} = F(X_t, U_t, N_t)$; E2a–d the boxed four-equation system; E3 softmax attention without temperature; E4 the EMA memory $H_{t+1} = (1-\rho)H_t + \rho \sum_i a_{i,t} V(y_{i,t})$; E5 the additive coupling $X_{t+1} = f_E(\cdot) + B_E \Pi_E(H_t)$; E6 the grammar tuple $\mathcal{L}_{\text{IWME}} = (\Sigma, G, \mu, \mathcal{A}, \mathcal{M}, \mathcal{T})$; E7–E9 H_ϵ, D_1, D_q; E10–E11 the dyadic cascade and $H_n = n\, h_2(p)$; E12 the cross-environment invariant $I(G_{E_1}) \approx I(G_{E_2})$; E13–E14 the IRT coordinate $w(\psi)$ and $X_{t+1} = F_E(\ldots, w_t)$; E15 the wake equation; E16 its memory-extended form with $\eta P_E(H_t, A_t)$; E17 the cycle chain $Y_t \to A_t \to H_{t+1} \to X_{t+1} \to G_{t+1}$; E18 the presence measure R (a conditional mutual information); E19 the model comparison M_0, M_1, Δ_{IWME}.

Experiments. X1 attention-controlled pattern formation; X2 relational history reconstruction; X3 cross-environment translation; X4 scale-dependent entropic grammar. None carries a full pre-registration (governing equations, units, boundary data, data source, budget, randomization, outcomes, inference plan, effect sizes, falsification thresholds, artifacts).

4 Phase 2: audit of Draft 1.0

Severity: CRITICAL (prevents a coherent formal theory), MAJOR (invalidates a stated claim or test design), MINOR (notation or wording). Reviewer perspectives: [M] mathematics, [P] physics, [C] computation, [S] statistics, [R] adversarial reviewer, [E] editor.

4.1 Architecture

F1 (critical, M/P). The observation map and the physical coupling share one symbol. D3 defines $\Pi_{E,\lambda}$ as the observer's representation of memory (hypothesis H1 of draft §7); E2c and E5 feed $\Pi_E H_t$ into the physical dynamics (hypothesis H2). The draft insists these differ and its equations merge them. Repair: an actuation map $C_E : \mathcal{H} \to \mathcal{U}_E$ distinct from $\Pi_{E,\lambda}$; H1 is the sub-hypothesis $C_E \equiv 0$.

F2 (critical, M). The recursive cycle is asserted and never written. Y_t is a free exogenous input, so H_t filters an external signal and the loop is open. Repair: $Y_t = \mathcal{Y}_E(X_t, Y_t^{\text{ext}}, N_t^y)$.

F3–F4 (major). The geometric output is defined three ways (D4, E2d, §5), and E2c and E17 disagree on whether X_{t+1} reads H_t or H_{t+1}. A physical observable can depend on H only through X; an output that reads H directly is a representation. Repair: $G_{t,\lambda}^{\text{phys}} = T_{E,\lambda}(X_t)$ and $G_{t,\lambda}^{\text{rep}} = T'_{E,\lambda}(X_t, H_t)$, one causal order (Section 6.1).

F5 (major, M/S). No probability structure. Mutual information (draft §7.2) and likelihood (§9) need one. The minimal formalism is a controlled Markov kernel on $Z_t = (X_t, H_t)$ with emission kernels for Y_t and G_t, a controlled POMDP.

F6–F8 (major/minor, E/P). "Intra-dimensional" is undefined; a candidate reading (within one representational perspective $\Pi_{E,\lambda}$, with no additional physical dimensions) awaits confirmation (Q1). Symbol collisions: λ is both scale and damping (damping becomes γ here), H is both memory and entropy (entropy becomes $\mathrm{H}[\cdot]$), M is both operator and model. E_t should hold slow parameters and boundary data only; τ_E needs a concrete map or removal.

4.2 Attention and memory

F9 (major, M/R; amended per correction (a)). E4 is a first-order IIR filter whose input is a convex combination of values. *For an attention sequence fixed in advance*, retention has time constant $\tau_{\text{mem}} = -1/\ln(1-\rho)$ set by ρ alone and attention selects content only (Proposition 6). *When the scores read the memory*, as in any architecture where $q = q(H_t)$, the map is a closed-loop recurrence whose Jacobian contains $(\rho/T)\,\mathrm{Cov}_a(v, k)$, and attention changes retention (Proposition 7): at $T = 1.2$, $\rho = 0.5$, the sensitivity of H_{10} to H_0 is 0.365 in closed loop against $(1-\rho)^{10} = 9.8 \times 10^{-4}$ in open loop (V02). The earlier conclusion that attention only selects content therefore holds for the fixed-input filter and fails for the feedback system. A forget gate $\rho_t = \rho(A_t)$ remains the way to make retention attention-dependent in open loop (V14).

F10 (major). E3 has no temperature and no score function. Ablation needs a continuous control: $s_{i,t} = \langle q(H_t), k(y_{i,t})\rangle / T$, with $T \to \infty$ uniform and $T \to 0$ hard selection; q, k, V must be fixed before any experiment.

F11–F12 (constructive). Attention sharpness is a bifurcation parameter (Section 6.4). The softmax is a Gibbs distribution with energies $-s_i$, so a physical attention operator has a standard reading: Boltzmann weighting of competing modes by growth rate, gain or overlap, with saturation supplying normalization; this is pattern selection by mode competition [28]. It is the natural candidate pathway for H2, and it is ordinary physics.

4.3 Physical coupling

F13 (major, P). $B_E \Pi_E(H_t)$ must carry the units of X, so B_E is dimensionful. A memory-dependent additive term in a conservative system does work and needs an energy source; E5 is a stateful nonlinear controller with an EMA/softmax structure. Novelty has to be located elsewhere.

4.4 Entropy as a fractal language

F14–F16 (major/minor). The grammar tuple lacks domains and mixes layers. With a finite alphabet Σ, a constraint language $G \subseteq \Sigma^*$, a measure μ on $\Sigma^{\mathbb{N}}$ and a symbolization $\sigma : X \to \Sigma$, the construction is symbolic dynamics with the thermodynamic formalism [12, 13, 10]; with a generating partition the entropy rate equals the Kolmogorov–Sinai entropy. The worked cascade is correct once the dyadic assumption $\epsilon = 2^{-n}$ is stated: $D_q = \log_2(p^q + (1-p)^q)/(1-q)$ exactly at every level. All eight hand-computed values of the Phase 0–2 table reproduce (V15), box counting on the exact level-16 measure reproduces the closed form to 2×10^{-16} (V16), and $H_n = n\, h_2(p)$ holds exactly (V17). The measure is exactly self-similar, so the draft's phrase "increasingly structured across scales" is incorrect. The D_q spectrum is a necessary condition for a shared mechanism and far from a complete invariant. This material is a baseline [8, 9, 11].

4.5 History dependence and identifiability

F18 (critical, M/S/R). "Future expression may depend upon relational history not completely recoverable from its immediately observed configuration" is trivial if "configuration" means the measured state (every hidden-state system) and unfalsifiable if it means the complete physical state (Proposition 5). The testable form is parametric: a specific low-complexity attention–memory class predicts held-out behaviour better than hidden-state models of matched dimension and information access.

F19–F21 (major). M_1 conditions on H_t, A_t, which are functions of $Y_{\le t}$, while M_0 is denied $Y_{\le t}$; any positive Δ_{IWME} is then partly an information-access artifact. Only the observable subspace of H (Kalman decomposition [16]) is identifiable. The invariant of E12 must be pre-registered as invariant under a declared transformation group, with known universality classes [27, 28] as the null.

4.6 The IRT history coordinate

F22–F23 (minor/major). $\kappa(s)$ is a scalar function along a path and should be called one. w is an integrator: if $\kappa = \kappa(\psi)$, w is a function of ψ and carries no information beyond ψ; with κ driven by

unobserved quantities, E13 is an accumulator with a pole at 1, against the EMA pole at $1-\rho$. The two are distinguishable after decoupling (Section 6.3).

F24 (corrected per (b)). The Phase 0–2 report warned that treating w as an independent coordinate while defining $w = w(\psi)$ would make the induced metric degenerate. Source C's Definition 2.1 does something different: three parameters (θ, φ, ψ), four ambient coordinates (x, y, z, w), $w = -\int_0^\psi \kappa$. The 4×3 Jacobian has rank three away from explicitly located circles, so the induced metric is nondegenerate there (Proposition 2, V25). The warning applies only to Source C's Definition 5.1 read as a four-coordinate line element on a three-manifold. The actual defect is the published metric, recomputed in Section 5.1.

F25 (standing position). An exponentially contracting coordinate is a reparametrization and absorbs no energy; an inverse reconstruction operator that uses later data is smoothing, which is inference in time and involves no retrocausality.

4.7 Wake-field dynamics

F26–F29 (major). Nothing in E15 is physically identified; a consistent unit assignment exists once $[W]$ is fixed, and S requires an explicit coarse-graining with its own scale. E16 places a discrete-time memory inside a continuous-time PDE; it needs a zero-order hold $P_E(H_t, A_t)(x) = \sum_j (H_t)_j \varphi_j(x)$ on $[t\Delta, (t+1)\Delta)$ or a continuous-time memory (implemented as `HybridWake`). E15 is the damped wave equation with sources; its energy identity and a second-order scheme are in Section 6.6. A nonzero inferred η is indistinguishable from an unmodeled source unless P_E has a pre-registered form.

4.8 The presence criterion

F31 (major, S/R). E18 is conditional transfer entropy [18], Granger causality in information form [17]. It is positive under any coupling, common cause or incomplete conditioning, so it measures coupling. The strengthened criterion, persistence after decoupling, is developed and computed in Section 6.3.

4.9 Experimental program and citations

F32–F35. The experiments lack items 4–16 of a pre-registration; X1's "matched conventional controllers" is undefined while the IWME model is itself a controller; X2's success criterion is met by any latent-state model; X3 and X4 lack pre-registered invariants, symbolization and scale range. The preamble and abstract cite Source D's "convergence conditions and attention measurements" and the body uses none (Section 5.3 shows Source D supplies none usable). References 4–8 are correctly cited; the primary sources on which the claims depend are added to the bibliography.

5 Source audits

5.1 Source C: the Inverse Resonant Toroid, recomputed

Source C, Definition 2.1, parametrizes $\mathcal{T}^{\mathrm{IRT}} \subset \mathbb{R}^4$ by

$$X(\theta, \varphi, \psi) = \big(\rho\cos\theta,\ \rho\sin\theta,\ r\sin\varphi\,\sinh\beta\psi,\ w(\psi)\big), \tag{1}$$

where $\rho = u\,e^{-\alpha\psi}$, $u = R - r\cos\varphi$ and $w = -\int_0^\psi \kappa$, with $R > r > 0$, $\alpha, \beta > 0$, $\psi \geq 0$. The computations use $(R, r, \alpha, \beta) = (3, 1, 0.3, 0.5)$ and two kernels: the constant $\kappa \equiv 1$ and the single mode of Source C's eq. (15), $\kappa(\psi) = e^{-0.2\psi}\cos 2\psi$.

Proposition 1 (Induced metric; L0)**.** *In the order* (θ, φ, ψ)*, with* $g_{\theta\varphi} = g_{\theta\psi} = 0$*,*

$$\begin{aligned}
g_{\theta\theta} &= u^2 e^{-2\alpha\psi},\\
g_{\varphi\varphi} &= r^2\big(\sin^2\varphi\, e^{-2\alpha\psi} + \cos^2\varphi\,\sinh^2\beta\psi\big),\\
g_{\varphi\psi} &= r\sin\varphi\,\big(\beta r\cos\varphi\,\sinh\beta\psi\cosh\beta\psi - \alpha u\, e^{-2\alpha\psi}\big),\\
g_{\psi\psi} &= \alpha^2 u^2 e^{-2\alpha\psi} + \beta^2 r^2\sin^2\varphi\,\cosh^2\beta\psi + \kappa(\psi)^2,\\
\det g &= u^2 e^{-2\alpha\psi}\big[r^2 e^{-2\alpha\psi} D^2 + \kappa^2 g_{\varphi\varphi}\big],
\end{aligned}$$

with $D = \beta r\sin^2\varphi\cosh\beta\psi + \alpha u\cos\varphi\sinh\beta\psi$*.*

Proof. $X_\theta = \rho\,\hat{e}_\theta$ is orthogonal to $X_\varphi = (\rho_\varphi\hat{e}_\rho,\ z_\varphi,\ 0)$ and $X_\psi = (\rho_\psi\hat{e}_\rho,\ z_\psi,\ -\kappa)$, which gives the θ row. The (φ, ψ) block is the Gram matrix of X_φ, X_ψ; its determinant is, by Cauchy–Binet, the sum of the squared 2×2 minors in the frame $(\hat{e}_\rho, \hat{e}_z, \hat{e}_w)$: $(\rho_\varphi z_\psi - z_\varphi\rho_\psi)^2 = r^2 e^{-2\alpha\psi} D^2$, $(\rho_\varphi\kappa)^2$ and $(z_\varphi\kappa)^2$, and the last two sum to $\kappa^2 g_{\varphi\varphi}$. A symbolic computation confirms every component and the determinant (V23); the closed form agrees with $J^{\mathsf{T}}J$ to 2×10^{-15} (V22). □

Proposition 2 (Rank; L0)**.** rank $J = 3$ *except on the circles (all* θ*) where* $\det g = 0$*:*

$$\{\psi = 0,\ \sin\varphi = 0\}\ \cup\ \{\kappa(\psi) = 0,\ D(\varphi, \psi) = 0\}.$$

Proof. Both bracketed terms of $\det g$ are nonnegative and $u > 0$. $g_{\varphi\varphi} = 0$ forces $\sin\varphi = 0$ and $\cos\varphi\sinh\beta\psi = 0$, hence $\psi = 0$, where $D = 0$ as well. Otherwise $\det g = 0$ needs $\kappa = 0$ and $D = 0$. □

For $\kappa \equiv 1$ only the two boundary folds at $\psi = 0$ remain (the $\psi = 0$ slice lies in the plane $z = 0$, which folds the φ-circle). For $\kappa = e^{-0.2\psi}\cos 2\psi$ there are two further circles on each zero $\psi_k = (2k+1)\pi/4$ of κ; on $(0, 6]$ these are at $\varphi \approx (2.3049, 3.9783)$, $(2.0483, 4.2349)$, $(2.0025, 4.2807)$, $(1.9932, 4.2900)$. The numerical rank is 2 and $\det g < 10^{-28}$ on all of them and 3 at generic points (V25). This confirms correction (b).

The published metric. Source C's Appendix A.1 prints $g_{\varphi\varphi} = r^2 e^{-2\alpha\psi}\cosh^2\beta\psi$, a $\sinh^2$ in place of $\cosh^2$ in $g_{\psi\psi}$, and $g_{\varphi\psi} = 0$. The $g_{\psi\psi}$ error has the closed form $g^{\mathrm{true}}_{\psi\psi} - g^{\mathrm{A.1}}_{\psi\psi} = \beta^2 r^2\sin^2\varphi$. At $(\varphi, \psi) = (2.5, 1.7)$ the relative errors are 3.6% in $g_{\varphi\varphi}$ and 8.0% in $g_{\psi\psi}$, and the omitted mixed component is $g_{\varphi\psi} = -0.563$ (V24); $g_{\theta\theta}$ and the vanishing θ-row are right.

Christoffel symbols. With the non-diagonal block, $\Gamma^a_{ij} = g^{ab}\Gamma_{b,ij}$ for $a, b \in \{\varphi, \psi\}$ and $\Gamma_{b,ij} = \frac{1}{2}(\partial_i g_{jb} + \partial_j g_{ib} - \partial_b g_{ij})$; in the θ row, $\Gamma^\theta_{\theta\varphi} = r\sin\varphi/u$ and $\Gamma^\theta_{\theta\psi} = -\alpha$. Source C's eq. (28) gets $\Gamma^\theta_{\theta\psi}$ right and omits $\Gamma^\theta_{\theta\varphi}$. Its other entries fail even against its own diagonal metric: the printed $\Gamma^\psi_{\theta\theta} = -\alpha u^2 e^{-2\alpha\psi}/g_{\psi\psi}$ has the wrong sign (-1.278 printed, $+1.278$ from eq. (27), at $(\varphi, \psi) = (0.9, 1.2)$), and the printed $\Gamma^\psi_{\varphi\varphi}$ is $+0.358$ against $+0.033$ (V31). The values from the correct metric at that point are $\Gamma^\psi_{\theta\theta} = +0.831$, $\Gamma^\psi_{\varphi\varphi} = -0.562$, $\Gamma^\varphi_{\varphi\psi} = +0.189$ (printed -0.031).

Proposition 3 (Scalar curvature; L0). *Let Σ be the profile surface $(\varphi, \psi) \mapsto (\rho, z, w) \in \mathbb{R}^3$ with metric g_Σ (the (φ, ψ) block). Then M is the warped product $\Sigma \times_\rho S^1$ and*

$$\mathrm{Scal} = 2K_\Sigma - 2\,\frac{\Delta_\Sigma \rho}{\rho}, \qquad K_\Sigma = \frac{\det \mathrm{II}_\Sigma}{\det g_\Sigma}.$$

Scal *depends on κ only through κ and κ'.*

Proof. $g = g_\Sigma + \rho^2 d\theta^2$, and for a warped product with a one-dimensional flat fibre $\mathrm{Scal} = \mathrm{Scal}_\Sigma - 2\Delta_\Sigma\rho/\rho$ [31]. For the second claim, M is a hypersurface of flat $\mathbb{R}^4$, so the Gauss equation gives $\mathrm{Scal} = (\operatorname{tr} S)^2 - \operatorname{tr} S^2$ with $S = g^{-1}\mathrm{II}$ [32]; II involves X and its second derivatives, and $X_{\psi\psi}$ carries $-\kappa'$ and no higher derivative. □

Three independent numerical routes, the Gauss equation with exact derivatives, the intrinsic Riemann tensor from finite differences of the metric, and the warped-product formula, agree to 3×10^{-6} (relative) at ten test points (V26). Changing $\kappa''(\psi_0)$ from 0 to 14 with $\kappa(\psi_0), \kappa'(\psi_0)$ fixed leaves $\mathrm{Scal}(\psi_0)$ unchanged to 2×10^{-8} (V27).

Proposition 4 (Sign; L0). *For constant $\kappa \equiv \kappa_0$, $\mathrm{Scal} \to 2K_{T_\psi}$ as $\kappa_0 \to \infty$, where T_ψ is the slice $\psi = \mathrm{const}$, a torus of revolution with elliptical profile and*

$$K_{T_\psi}(\varphi) = -\frac{r^3 \sinh^2\beta\psi\, \cos\varphi}{u\, g_{\varphi\varphi}^2}.$$

Hence Scal *takes both signs for all large κ_0; $\int_{T_\psi} K\, dA = 0$ by Gauss–Bonnet on every slice with $\psi > 0$.*

Numerically the error $|\mathrm{Scal} - 2K_{T_\psi}|/|2K_{T_\psi}|$ is 1.6×10^{-1}, 1.6×10^{-3}, 1.6×10^{-5} at $\kappa_0 = 10, 100, 1000$, consistent with $O(\kappa_0^{-2})$, and $K_{T_\psi} < 0$ exactly where $\cos\varphi > 0$ (V28). At $\kappa_0 = 1$ the curvature is negative on 67% and positive on 33% of the chart $(\varphi, \psi) \in [0, 2\pi) \times [0.05, 4]$; for the damped kernel the split is 56%/44% (V29, Figure 1). Source C's conclusion "negative curvature confirms inward-focusing geometry" is false, and its eq. (29) disagrees with the curvature in value and in sign (sign differs at 2 of 5 test points; e.g. $+2.736$ against -8.186, V30); it also contains κ''/κ, which Proposition 3 excludes, and $3\beta^2/\cosh^2\beta\psi$, which lacks the inverse-length-squared units of a scalar curvature.

Shadow reconstruction. Source C's eq. (16) is offered as the integral of eq. (15) and is incorrect: it integrates the exponential and the cosine factors separately. Its derivative misses $-\kappa$ by up to 2.92 on $[0.1, 3]$ and it gives $w(0) = -5 \neq 0$; the exact antiderivative $-\sum_n A_n \left[e^{-\lambda_n\tau}(\omega_n \sin(\omega_n\tau + \phi_n) - \lambda_n \cos(\omega_n\tau + \phi_n))/(\lambda_n^2 + \omega_n^2)\right]_0^\psi$ is implemented (V32).

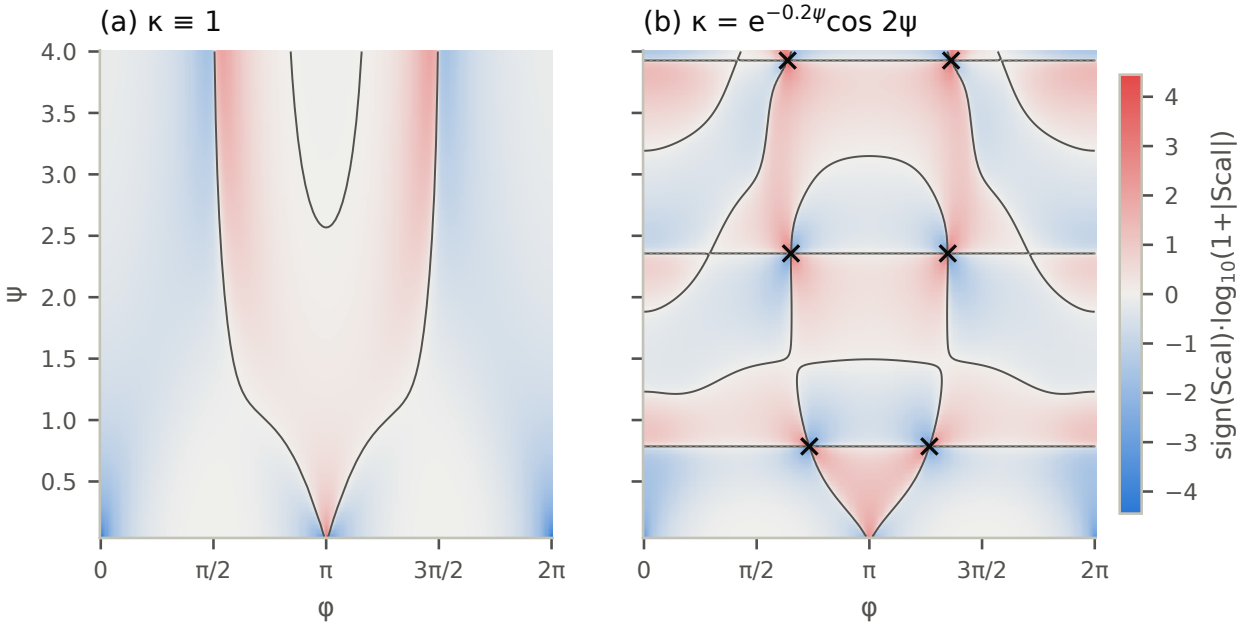


Figure 1: Scalar curvature of the IRT over the chart, signed logarithmic scale (blue negative, red positive, dark contour Scal = 0). (a) $\kappa \equiv 1$. (b) $\kappa = e^{-0.2\psi}\cos 2\psi$; dotted lines mark the zeros of κ and crosses the rank-2 circles of Proposition 2. Both signs occur on sets of large measure.

Definition 5.1. Source C also writes $ds^2 = g_{\mu\nu}dx^\mu dx^\nu + h(\psi)\,dw^2$ with $h = e^{2\beta\psi}$ and $g_{\mu\nu}$ "the standard toroidal metric". Read on the 3-manifold, $dw = -\kappa\,d\psi$ and the line element is a third metric, with $e^{2\beta\psi}\kappa^2$ in place of κ^2; read as a metric in four coordinates on a three-manifold it is degenerate, which is the case the original F24 described. The source needs one metric. The induced metric of Proposition 1 is the one consistent with Definition 2.1.

Consequences for IWME. The history coordinate w remains an integrator (F23). If the IRT stays in the program, its geometry is that of Propositions 1 and 3; its published curvature claims, reception principle and "higher-dimensional intelligence" interpretation are L3 and play no role in any test below.

5.2 Source B: the Emergent Fractal Law and the wake system

Categorical content. Several statements are ill-typed as written. Axiom 1.3 asks for a profunctor V "mediating $\pi\tau \Rightarrow \tau\pi$" with $\tau : \mathcal{S} \to \mathcal{S}_{\leq m}$, $\pi : \mathcal{S}_{\leq m} \to \mathcal{S}$; $\pi\tau$ is an endofunctor of $\mathcal{S}$ and $\tau\pi$ an endofunctor of $\mathcal{S}_{\leq m}$, so no natural transformation between them exists. Definition 2.1 (Fubini–Stokes) has the same defect. $[\tau, \delta] \simeq 0$ commutes a functor with a coboundary operator on cochains, which needs a chain-level action of τ that is never given. Theorem 1.1 cites Axioms A–D, which are numbered 1.1–1.4. The model example asserts that formality obstructions of A_∞-algebras become exact at a finite stage; non-formal algebras exist (formality forces all Massey products to vanish [33], and the complement of the Borromean rings carries a nonvanishing triple Massey product), so Axiom 1.2 fails in its own model. The effective dimension $D_{\text{eff}} = \limsup \log N(r) / \log \lambda^r$ with $N(r) \sim \lambda^r$ equals 1 identically (V21); a dimension needs the branching number and the scale factor to be independent, $D = \log b / \log \lambda$.

A finite model of the EFL fixed point. The Phase 0–2 report asked for a finite-dimensional model before any higher-categorical claim. Take measures on $[0,1]$, Φ_λ the Hutchinson operator $(\Phi\mu)(A) = p\,\mu(2A) + (1-p)\,\mu(2A-1)$. It contracts the Wasserstein-1 distance by exactly $\frac{1}{2}$ (the two branches do not overlap), its unique fixed point is the binomial measure, and $\Phi(\text{binomial}_n) = \text{binomial}_{n+1}$; the measured ratio is $\frac{1}{2}$ to 2×10^{-14} (V20) [14]. This realizes "$E \simeq \Phi_\lambda E$" as a theorem, with the colimit replaced by the limit of iterates.

The PDE system (4.2). With damping renamed γ:

(B1) *Defect identity (Prop. 4.2), false.* For a vortex of charge m, $\nabla \times \nabla\theta = 2\pi m\,\delta$ and $\Delta\theta = \nabla \cdot \nabla\theta = 0$ in distributions [29]. On a lattice, the circulation around every square loop enclosing the core is $2\pi m$ for $m \in \{1, -1, 2\}$ (loops of half-width ≥ 2), and the flux of $\nabla\theta$ through each loop, the integral of $\Delta\theta$ inside it, is 0 to machine precision; Prop. 4.2 predicts $2\pi m$ (V36). Sources for $\mathcal{W}$ must be built from the vorticity of the phase.

(B2) *Coherence (4.2c).* $\partial_t c + \mu c = -\gamma_c |\nabla \arg\psi|^2$ has the steady state $-\gamma_c |\nabla\theta|^2/\mu < 0$; with $c_0 = 1$, $\mu = 1$, $\gamma_c |\nabla\theta|^2 = 0.5$ it crosses 0 at $t = \ln 3$ (V37), and near a vortex $c^* = -\gamma_c m^2/(\mu r^2) \to -\infty$. The stated range $c \in [0,1]$ is not invariant; a bounded coherence needs a nonlinear self-term such as $\mu c(1-c)$.

(B3) *Entropy (4.2b).* $\partial_t S - D_S \Delta S = -\nabla \cdot J_S$ with $J_S = -\kappa \nabla S$ is diffusion with coefficient $D_S + \kappa$; the flux term duplicates D_S. Signs are consistent and the problem is well posed; with $\partial_n S = 0$ the total $\int S$ is conserved and there is no production term.

(B4) *Energy (Prop. 4.1).* The conclusion "if S and c are stationary, $\mathcal{E}(t)$ decays" needs zero forcing: with a stationary nonzero $\alpha\Delta S$ and the wake at rest, the energy rises from 5.6×10^{-8} to 3.6×10^{-5} (V38).

(B5) *Well-posedness (Thm. 4.1).* Point sources $q_k \delta(x - x_k)$ lie outside H^{-1} for $d \geq 2$: the 2D static response $-\ln r/2\pi$ has $\int_{h<r<1} |\nabla W|^2 = \ln(1/h)/2\pi \to \infty$ (V42), so the claimed $C([0,T]; H^1)$ solution does not exist without regularized cores.

(B6) *Variational principle.* The Euler–Lagrange equation of (4.4) is $W_{tt} - v^2 W_{xx} + v^2\omega^2 W = \chi v^2 S_{xx} + v^2 \xi(1-c)$: it has a restoring term absent from (4.2a), and its coherence source is $(1-c)$ where (4.2a) has $\partial_t(1-c)$ (V41).

(B7) *Worked examples.* The localized steady wake (4.9) requires a mass term $\gamma_m W$; with (4.2a) as written the steady response to a point source grows with the domain, $W(0) = L/2$ for $L = 1, 2, 4$ (V39). For the decoherence pulse the uniform mode reaches $W(\infty) = \beta/\gamma$ for every σ (2.000000 at $\sigma = 2$ and 0.5); Source B states $\beta\sigma/\gamma$ (V40).

(B8) *Scheme (4.13).* It discretizes the damping with a backward difference: first order in time (observed order 1.005, against 1.998 for centered damping, V34), and stable only for $v\Delta t/\Delta x < \sqrt{(1-\gamma\Delta t/2)/d}$ with $\gamma\Delta t < 2$. The centered scheme of Section 6.6 is stable up to $1/\sqrt{d}$ whatever $\gamma\Delta t$ (V33, Figure 5a).

AI instantiation (4.10). $\text{PC}(E) = (\max_i w_i - 1/M)/(1 - 1/M)$ with w the principal eigenvector of EE^T scores M identical answers 0, mutually orthogonal answers 1, and is undefined for an antipodal pair (the eigenvector changes sign and $\sum w_i = 0$) (V46). `kgirl.main.spectral_coherence`

already replaces it, on its fallback path, by the rescaled principal share $(\lambda_{\max}/M - 1/M)/(1 - 1/M)$, which scores agreement correctly and is sign-blind (an antipodal pair scores 1). The rescaled squared mean resultant length $(\|\bar{e}\|^2 - 1/M)/(1 - 1/M)$, the order parameter of unit vectors [26], scores the three cases $1, 0, 0$ and is the sign-aware choice. `PhaseCoherence.scalar` in `kgirl.main` still applies (4.10) when its optional topological backend is present.

5.3 Source D: The Convergence Point

Draft 1.0 cites Source D for "convergence conditions and attention measurements". Source D's attention condition is relational and qualitative, and its quantitative content is the WR-039T pipeline. The checks:

- *Tier 1.* The reported Walsh coefficients of $[1, 1, 1, -1, 1, 1, 1, 1]$ are $[4, 0, 0, -4, 0, 4, 4, 0]$. The transform is $[6, 2, 2, -2, -2, 2, 2, -2]$ in natural order; no ordering or normalization yields the reported vector, which is the transform of $[1, 1, 1, -1, -1, 1, 1, 1]$. Both vectors have energy $64 = N \cdot E_{\text{in}}$, so the reported Parseval PASS cannot detect the error (V43): Parseval holds for every orthogonal transform of every signal and verifies the implementation of the transform only.
- *Tier 2.* $\varphi(a^k) = \varphi(a)^k$ for $\varphi =$ reduction mod 3 on $\mathbb{Z}_9$ holds for all a, k (0 failures in 279) because 3 divides 9; a check that cannot fail carries no information. No unital ring map $\mathbb{Z}_4 \to \mathbb{Z}_3$ exists (V44). PAEF scores are reported without a formula.
- *Prediction 5.* "Energy concentrated in low-order Walsh coefficients" is maximal (share 1.000) for a constant signal and 0.011 for chaotic noise (V45): it measures smoothness, the reverse of the complexity it is meant to index.
- *Tier 8 and the appendix.* One 100-iteration run per system, 546.3 → 543.5 ms (−0.51%) with no dispersion, cannot separate system differences from serving-side variation (context length, caching, batching, sampling). The TRIBE v2 appendix compares model-predicted cortical responses to single sentences without statistics; the audio-only versus trimodal contrast follows from which encoder sees the words.

Source D therefore provides no attention measurement IWME can use. It can be cited for the conceptual claim that attention is a relational condition; the abstract's dependency should be removed or rephrased.

5.4 Source E: scope note

From Triangle to Form (E8 holography, the proton radius) is outside IWME's dependency graph and is not audited here. One methodological point transfers: its automated checks, such as $\sqrt{12/\cos^2 30^\circ} = 4$, verify arithmetic identities that hold by construction; like Source D's Parseval check, they test the code and say nothing about the physical identifications.

6 Phase 3: formal rehabilitation

6.1 Repaired core

One causal order, the loop closed, observation separated from actuation:

$$\begin{aligned} Y_t &= \mathcal{Y}_E\big(X_t,\, Y_t^{\mathrm{ext}},\, N_t^y\big), & A_t &= \mathcal{A}(H_t, Y_t, E_t) \in \Delta^{n-1}, \\ H_{t+1} &= \mathcal{M}\big(H_t, A_t, Y_t, N_t^h\big), & X_{t+1} &= F_E\big(X_t, U_t, N_t^x,\, C_E(H_t)\big), \\ G_{t,\lambda}^{\mathrm{phys}} &= T_{E,\lambda}(X_t), & G_{t,\lambda}^{\mathrm{rep}} &= T'_{E,\lambda}(X_t, H_t). \end{aligned} \tag{2}$$

E_t holds parameters and boundary data; $\Pi_{E,\lambda}$ enters only through T'; H1 is $C_E \equiv 0$ and H2 is $C_E \neq 0$. With the noises independent, $Z_t = (X_t, H_t)$ is a controlled Markov chain and (Y_t, G_t) are its emissions (a controlled POMDP). *Emission* is then the map $\mathcal{M} \circ (\mathcal{A}, \mathcal{Y}_E)$ together with C_E: what memory writes and what it sends to the physics. *Intra-dimensional* remains the candidate reading of F6 until confirmed (Q1). The implementation is `kgirl.iwme.core.IWMECore`.

Proposition 5 (Observational equivalence; L0)**.** *The law of $(Y_t, G_t)_{t\geq 0}$ given (U_t, E_t) generated by (2) is generated by a single-state model $Z_{t+1} = F'(Z_t, U_t, E_t, N_t)$, $Y_t = \mathcal{Y}'(Z_t, N_t^y)$, $G_t = T'(Z_t)$ with no distinguished memory variable.*

Proof. With $Y = \mathcal{Y}_E(X, Y^{\mathrm{ext}}, N^y)$, set

$$F'(Z, \cdot) = \Big(F_E\big(X, U, N^x, C_E(H)\big),\ \mathcal{M}\big(H, \mathcal{A}(H, Y, E), Y, N^h\big)\Big), \qquad T'(Z) = T(X).$$

Substitution reproduces (2). The unit test `test_augmented_state_reproduces_observables` checks the implementation on a nonlinear instance. □

No statistic of the observables separates relational memory from unmodeled physical state. Evidence for IWME can come from parametric restrictions on $(\mathcal{A}, \mathcal{M}, C_E)$ that beat matched hidden-state models, or from interventions on the memory substrate independent of X.

6.2 Hypothesis

The investigator's clarification fixes the status of the attention mechanism: the temperature-controlled bifurcation of Section 6.4 is established mathematics (the mean-field pitchfork of Curie–Weiss and Hopfield type [24]) and serves as the mechanism on which predictions are built. The working hypothesis for Phases 3–4, pending the formal answer to Q3, is the parametric one: *an attention–memory model class with temperature T and write rate ρ predicts held-out behaviour of a physical system after interaction better than hidden-state models of matched latent dimension and identical information access* $(X_{\leq t}, Y_{\leq t}, E_{\leq t}, U_{\leq t})$ (L2). The physical claim $C_E \neq 0$ is retained as the explicitly labeled extension M4 (L2–L3).

6.3 Persistence after decoupling

Interact A and B for $t < t_c$, sever the channel at t_c, estimate the physical state $\hat{X}_{t_c}^A$ as completely as the instruments allow, and define

$$R_{\mathrm{persist}}(\ell) = I\big(G_{t_c+\ell}^A;\, Y_{<t_c}^B \,\big|\, \hat{X}_{t_c}^A, U, E\big). \tag{3}$$

If the estimate were complete and the dynamics Markov, $R_{\text{persist}} = 0$. For the linear-Gaussian instance $H_{t+1} = \mu H_t + \rho Y_t^B$ ($\mu = 1-\rho$ EMA, $\mu = 1$ accumulator), $X_{t+1} = aX_t + cH_t + s_x\xi_t$, $G_t = X_t + s_g\zeta_t$, the conditional mutual information is computed exactly from covariances (`persistence_information`; $a = 0.8$, $\rho = 0.3$, $s_x = s_g = 0.1$, $t_c = 40$):

- under H1 ($c = 0$), $R_{\text{persist}} = 0$ exactly;
- with $c = 0.5$, the 30-step block carries 1.366 bits given X_{t_c};
- given (X_{t_c}, H_{t_c}) it drops to 10^{-14}: the criterion certifies state incompleteness, as Proposition 5 requires;
- the lag profile separates memory kinds: at lag 60 the EMA memory gives 1.5×10^{-10} bits and the accumulator (the pole of the IRT integrator w) 2.68 bits (V47, Figure 5b).

The parametric content of IWME enters through this profile: an EMA memory predicts geometric decay governed by $1-\rho$, and the attention model predicts how that decay changes with T (Section 6.4). A pre-registration skeleton:

Table 2: Protocol B (persistence after decoupling), pre-registration skeleton.

Item	Specification
Hypothesis	$R_{\text{persist}}(\ell) > 0$ after severing, with lag profile predicted by the attention–memory class at fitted (T, ρ) better than by matched hidden-state models (L2).
Arms	coupled then severed; never coupled (C_E arm control); severed with maximal state measurement (incompleteness control).
Estimand	$R_{\text{persist}}(\ell)$, $\ell = 1, \ldots, L$, with $\hat{X}_{t_c}$ covariance reported.
Models	M1 attention–EMA; M1′ attention with forget gate; M0 linear-Gaussian latent model of matched dimension; M0′ accumulator; all given identical inputs.
Outcome	held-out log-likelihood of $G_{t>t_c}$; primary contrast M1 versus best M0 at equal parameter count.
Falsification	M1 does not beat M0 by a pre-set margin, or the fitted lag profile is flat (accumulator-like) where M1 predicts decay.
Determinism	logistic-map variates (`ChaosRNG`), seeds listed; code `kgirl.iwme.core`.

6.4 Attention theory

Keys k_i, values v_i, scores $s_i = \langle H + b, k_i \rangle / T$ with an input bias b, weights $a = \text{softmax}(s)$, read $m(H) = \sum_i a_i v_i$, and $\Phi(H) = (1-\rho)H + \rho\, m(H)$. Let D_k, D_v be the diameters of the key and value sets and $C_a(H) = \sum_i a_i (v_i - m)(k_i - \bar{k})^{\mathsf{T}}$, $\bar{k} = \sum_i a_i k_i$.

Proposition 6 (P2, open loop; L0)**.** *If the read sequence m_s is fixed in advance, $H_t = (1-\rho)^t H_0 + \rho \sum_{s<t} (1-\rho)^{t-1-s} m_s$ and $\partial H_t / \partial H_{t-k} = (1-\rho)^k I$ for every attention sequence (V01).*

Proposition 7 (P2′, closed loop; L0)**.**

$$D\Phi(H) = (1-\rho)I + \frac{\rho}{T} C_a(H), \qquad \frac{\partial H_t}{\partial H_s} = \prod_{\tau=s}^{t-1} D\Phi(H_\tau) \quad \textit{(ordered)}.$$

Retention therefore depends on T and on the key–value geometry.

Proof. $\partial a_i/\partial H = a_i(k_i - \bar{k})/T$, so $Dm = T^{-1}\sum_i v_i a_i (k_i - \bar{k})^\mathsf{T} = C_a/T$ since $\sum_i a_i(k_i - \bar{k}) = 0$; the chain rule gives the product. □

At $H = 0$ in the two-point example below the retention time $-1/\ln \mu(0)$ is 41.5, 5.48, 2.47 steps at $T = 1.05, 1.5, 3$ and tends to $\tau_{\text{mem}} = 1.443$ as $T \to \infty$ (V03); along a trajectory at $T = 1.2$ the closed-loop sensitivity at lag 10 is 374 times the open-loop value (V02). Figure 3a shows the retention time at the stable state diverging at T_c from both sides.

Lemma 8 (Contraction; L0)**.** $\|C_a\|_{\text{op}} \le D_v D_k/4$ *for every weight vector* a*. Hence* $\text{Lip}(\Phi) \le 1 - \rho + \rho D_v D_k/(4T) < 1$ *when* $T > D_v D_k/4$*, and every memory trajectory is entrained by its inputs.*

Proof. For unit x, y, $x^\mathsf{T} C_a y = \text{Cov}_a(\langle x, v\rangle, \langle y, k\rangle) \le \sqrt{\text{Var}_a\langle x, v\rangle \; \text{Var}_a\langle y, k\rangle}$, and a distribution on an interval of length ℓ has variance at most $\ell^2/4$ (compare with the midpoint). The projections have ranges at most D_v and D_k. □

The bound is attained in the two-point example at $H = 0$, where $\|C_a\| = 1 = D_v D_k/4$ (V05); on random configurations the worst ratio is 0.958 (V04). This sharpens the Phase 0–2 bound $D_v D_k/T$, which gave $T > 4$ for an exact threshold of 1.

Proposition 9 (P3, pitchfork and critical slowing; L0)**.** *Take* $n = 2$, $d = 1$, $v = k = (+1, -1)$*, so* $m(H) = \tanh\big((H + b)/T\big)$*. At* $b = 0$*: for* $T > 1$ *the unique fixed point* 0 *is stable with multiplier* $1 - \rho + \rho/T$*; for* $T < 1$ *it is unstable and* $\pm H^*(T)$*,* $H^* = \tanh(H^*/T)$*, are stable with multipliers* $1 - \rho + \rho(1 - H^{*2})/T \in (1 - \rho, 1)$*;* $H^* \approx T\sqrt{3(1 - T)}$ *near* $T_c = 1$*. At* $T = 1$ *the fixed point is unique and attracting with*

$$H_t \simeq \Big(\frac{3}{2\rho t}\Big)^{1/2}.$$

Proof. The fixed-point and stability statements follow from the concavity of $\tanh(\cdot/T)$ on $H > 0$ and its slope at the crossing. At $T = 1$, $\Phi(H) = H - \rho(H - \tanh H) = H - \rho H^3/3 + O(H^5)$, so $H_{t+1}^{-2} = H_t^{-2} + 2\rho/3 + O(H_t^2)$. □

Checks: $H^*(0.5) = 0.957504$ with multipliers 0.583 and 1.5 (V06); $H^*/(T\sqrt{3(1 - T)}) = 1.000060$ at $T = 0.9999$ (V07); $H_t\sqrt{t} = 1.731982$ at $t = 10^5$ against $\sqrt{3} = 1.732051$ for $\rho = 0.5$ (V08); log–log slopes 0, −0.490, −7.49 at $T = 0.8, 1, 1.2$ (V09, Figure 2). In words: for $T < 1$ two stable nonzero equilibria appear; for $T \ge 1$ the unique equilibrium is zero, and convergence at the critical point is algebraic.

Proposition 10 (P3′, crossing type; L0)**.** *For centered values,* $H = 0$ *is a fixed point with* $D\Phi(0) = (1 - \rho)I + (\rho/T)C_0$*,* $C_0 = \text{Cov}_{\text{unif}}(v, k)$*, and eigenvalues* $\mu_j = 1 - \rho + \rho c_j/T$*. As* T *decreases, a real* $c > 0$ *crosses* $+1$ *at* $T = c$ *(pitchfork: multistable lock-in); a real* $c < 0$ *crosses* -1 *at* $T_f = \rho|c|/(2 - \rho)$ *(flip: period-2 memory); a complex pair crosses* $|\mu| = 1$ *(Neimark–Sacker: quasi-periodic memory) [25].*

For keys $\pm e_1, \pm e_2$ the thresholds are 0.5 (aligned values), 1/6 (anti-aligned) and $1/\sqrt{12} = 0.2887$ (values rotated by 90°), and the spectral radius crosses 1 at each (V13). For the anti-aligned two-point map $T_f = 1/3$ at $\rho = 0.5$ and a period-2 orbit of amplitude 0.127 exists at $0.95\,T_f$ (V12). Sharp attention therefore destabilizes memory when keys and values are anti-aligned: the counterexample requested in the Phase 0–2 priority list.

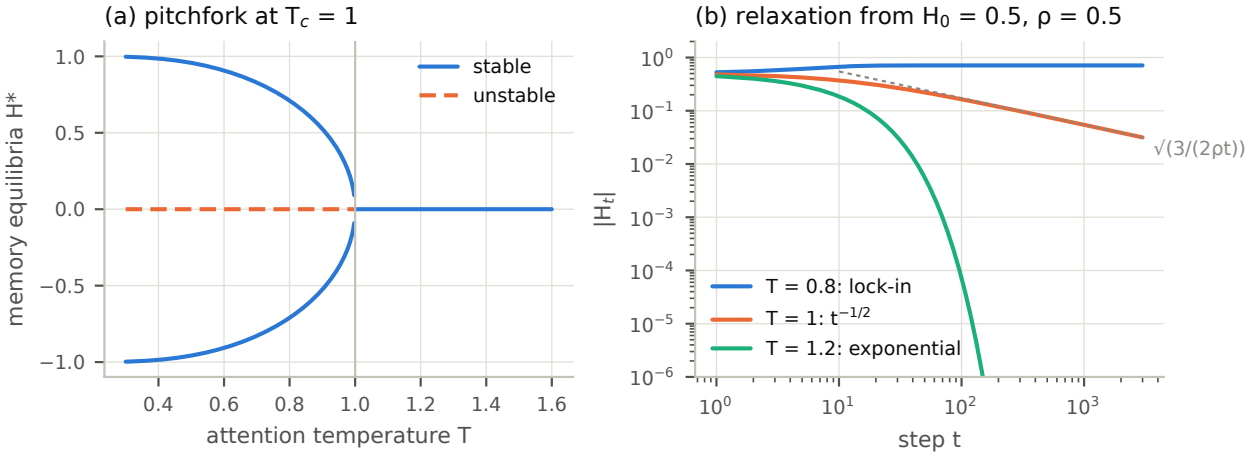


Figure 2: Two-point attention memory, $\rho = 0.5$. (a) Equilibria versus temperature: pitchfork at $T_c = 1$. (b) Relaxation of $|H_t|$: lock-in at $T = 0.8$, the algebraic law $(3/2\rho t)^{1/2}$ at $T = 1$, exponential decay at $T = 1.2$.

History dependence. With input bias b, the equilibria satisfy $b = T\,\mathrm{artanh}\,H - H$. For $T < 1$ the curve folds at $H = \pm\sqrt{1-T}$ and two stable branches coexist for $|b| < b_c = \sqrt{1-T} - T\,\mathrm{artanh}\,\sqrt{1-T} \approx \frac{2}{3}(1-T)^{3/2}$; for $T \geq 1$ the solution is unique. The adiabatic loop area is 0.658, 0.173, 0.0111 at $T = 0.6, 0.8, 0.95$ and 0 above T_c (V10); the measured exponent of $(1-T)$ is 1.975, against 2 from width $\times$ height (V11, Figure 3b). Finite-rate sweeps show small loops above T_c from critical slowing, so an experiment must extrapolate in sweep rate.

6.5 Multifractal module

For multipliers driven by a two-state Markov chain, $\tau(q) = (q-1)D_q = -\log_2 \rho(M_q)$ with $M_q(s,s') = \sum_c w_c(s)^q P(s' \mid s, c)$; the annealed partition function $\pi_0^{\mathsf{T}} M_q^n \mathbf{1}$ matches exact enumeration to 7×10^{-16}, and the i.i.d. limit returns the binomial value exactly (V18); for the reference chain $D_1 = 0.916$, $D_2 = 0.828$. For a continuous memory carried down the tree, $h' = (1-\rho)h \pm \rho$ with left share $\sigma((h+0.3)/0.8)$, the spectrum is the leading eigenvalue of a transfer operator: $\tau(2) = 0.49709$ (converged in the grid), against enumeration slopes 0.50360, 0.49916, 0.49815 for $n = 10{\to}18$, $14{\to}20$, $16{\to}22$, converging toward it (V19). The binomial cascade with the same $D_1 = 0.8222$ ($p = 0.7430$) has $D_2 = 0.694$; the memory cascade has $D_2 = 0.497$. Expectation E-A is met on both counts: memory changes the spectrum, and the change stays inside the thermodynamic formalism (Figure 4). A claim of novelty would need the attention parameterization to predict a measured spectrum from fewer parameters than a matched Markov cascade.

6.6 Wake module

For $W_{tt} + \gamma W_t - v^2 \Delta W = f$ on a periodic grid with the $(2d+1)$-point Laplacian, the centered scheme

$$(1 + \tfrac{\gamma\Delta t}{2})W^{n+1} = 2W^n - (1 - \tfrac{\gamma\Delta t}{2})W^{n-1} + \Delta t^2\big(v^2 \Delta_h W^n + f^n\big)$$

is second order (observed 1.998, V34) and stable for $v\Delta t/\Delta x < 1/\sqrt{d}$ for every $\gamma\Delta t$ (Jury conditions, confirmed by bisection over the discrete spectrum to 6×10^{-11}, V33) [30]. With $E^{n+\frac{1}{2}} = \frac{1}{2}\|(W^{n+1} -$

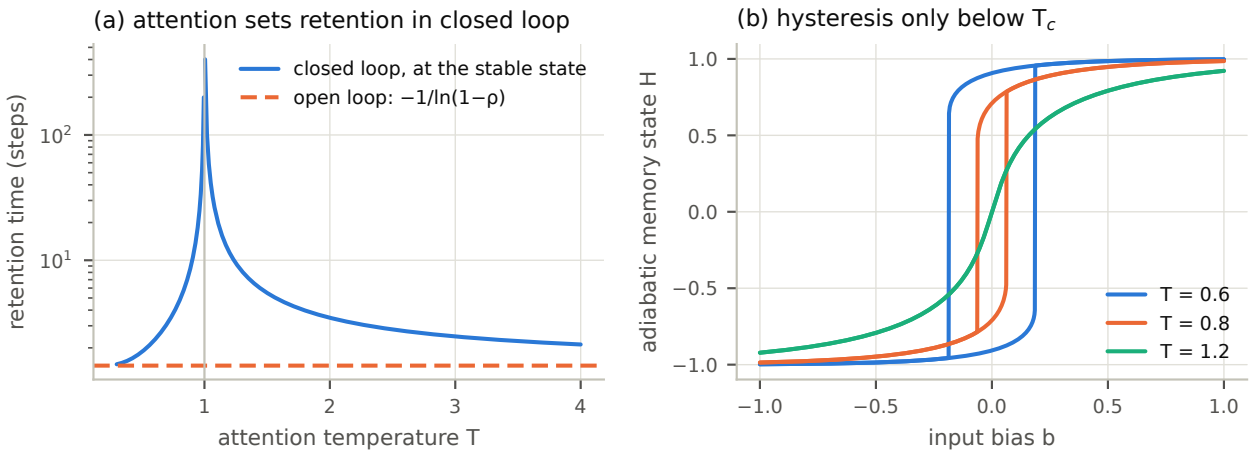


Figure 3: (a) Retention time at the stable equilibrium in closed loop (blue) against the open-loop value $-1/\ln(1-\rho)$ (orange): attention sets retention near T_c. (b) Adiabatic memory state under a slow sweep of the input bias: loops for $T < 1$, a single-valued response at $T = 1.2$.

$W^n)/\Delta t\|^2 + \frac{v^2}{2}\langle \nabla_h W^{n+1}, \nabla_h W^n\rangle$ it satisfies the exact identity

$$E^{n+\frac{1}{2}} - E^{n-\frac{1}{2}} = -\gamma\Delta t\left\|\frac{W^{n+1}-W^{n-1}}{2\Delta t}\right\|^2 + \Delta t\left\langle f^n, \frac{W^{n+1}-W^{n-1}}{2\Delta t}\right\rangle,$$

verified to 9×10^{-15} in a forced, damped 2D run (V35); the solver records work and dissipation as observability hooks. The memory couples through `HybridWake`, the zero-order hold of F27.

7 Revised claim ledger

Table 3: Claims of Draft 1.0 after Phases 2–3.

	Claim (draft §)	Classification	Note
C1	Four-equation system is a coherent state-space model (§3, §11)	construction, defective; repaired	F1–F5; (2)
C2	Softmax weights are a distribution (§3.1)	theorem (trivial)	—
C3	E4 implements selective memory integration (§3.1)	construction	content selection in open loop; retention also set by attention in closed loop (P2′)
C4	A memory-dependent physical term may exist (§3.2)	conjecture (H2), L2	a stateful controller unless a new coupling is specified
C5	$\mathcal{L}_{\mathrm{IWME}}$ is a language (§4)	construction, incomplete	symbolic dynamics after F14
C6	$H_n = n\,h_2(p)$ (§4.2)	theorem (standard)	V17

	Claim (draft §)	Classification	Note
C7	$D_1 \approx 0.8813$, $D_2 \approx 0.7859$ at $p = 0.7$	theorem (closed form)	V15; baseline
C8	A recurring D_q identifies related scaling (§4.1)	weak conjecture	necessary condition only
C9	Future expression depends on unrecoverable history (§5)	trivial or unfalsifiable	F18, Proposition 5
C10	Cross-environment invariant E12 (§5)	conjecture	F21
C11	IRT w as memory (§5.1)	integrator; geometry recomputed	Section 5.1; F23
C12	Wake equation models information disturbances (§6)	damped wave; physical identification unsupported	Sections 5.2 and 6.6
C13	Memory-coupled wake (§6)	construction after F27	`HybridWake`
C14	H1/H2 distinction (§7)	construction of hypotheses	retained as $C_E \equiv 0$ versus $C_E \neq 0$
C15	R measures presence (§7.2)	transfer entropy; "presence" L3	replaced by R_{persist}
C16	Δ_{IWME} criterion (§9)	construction	needs matched information access
C17	The draft draws on Source D's attention measurements	unsupported	Section 5.3
C18	Attention has a foundational operational role (§1.1, §7)	conjecture, sharpened	necessary for content selection, bistability and closed-loop retention; the persistence floor is set by ρ
C19	"This closes the recursive cycle" (§7.1)	unsupported as written; repaired	F2
C20	"Matter does not need to understand mathematics" (§11)	rhetorical, L3	move to Discussion

No claim of Draft 1.0 is empirically supported or independently replicated; this matches the draft's integrity note. Source claims found false or uninformative are listed in Section 5 and Appendix A.

8 Interrogation of the draft's questions, revised

1. *Is nonuniform selection necessary for structured memory?* For the persistence floor set by ρ, no; for content selection, bistability and attention-dependent retention, yes (P2, P2′, P3).
2. *Persistent memory with uniform attention?* Yes, with $\tau_{\text{mem}} = -1/\ln(1-\rho)$; in closed loop sharp attention moves the retention time between τ_{mem} and divergence at T_c (aligned values)

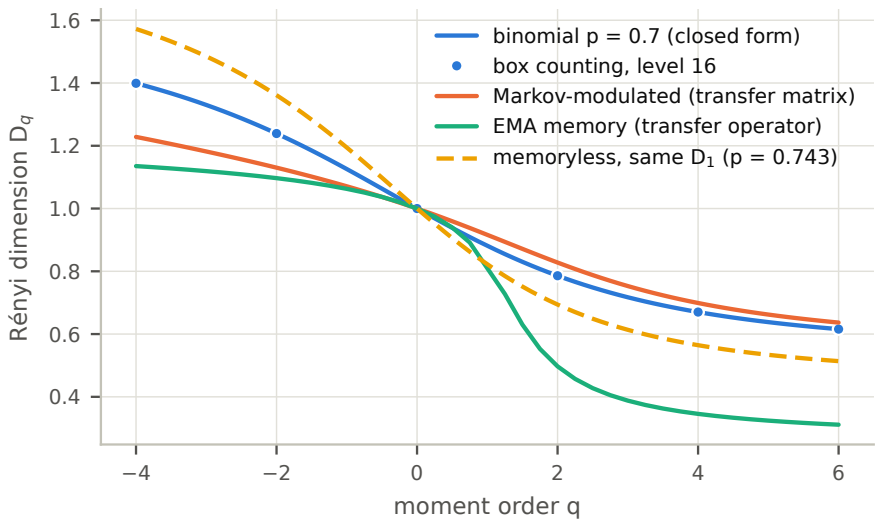


Figure 4: Rényi spectra. Binomial cascade (closed form and box counting), Markov-modulated cascade (transfer matrix), memory cascade (transfer operator), and the memoryless binomial with the memory cascade's D_1.

or shortens it (anti-aligned).

3. *Retention improved or suppressed?* Both occur in closed loop, depending on the spectrum of the key–value cross-covariance (P3′).
4. *Stable attractors from selective coupling?* Yes for $T < T_c$; none for $T > D_v D_k/4$ (Lemma 8).
5. *Entropy-rate change?* Memory changes the spectrum (D_2: 0.497 against 0.694 at matched D_1) through a transfer operator.
6. *Cross-scale preservation?* The memory cascade has a well-defined spectrum in the limit; finite-n corrections decay like $1/n$.
7. *History-dependent geometry?* Only through $C_E \neq 0$ (H2) or representation (H1); R_{persist} measures it after decoupling.
8. *Disappears under ablation?* Bistability, hysteresis and the attention-dependent excess retention, yes; the ρ-floor, no.
9. *Equivalent outcomes from conventional filters?* Persistence, yes (the EMA is an IIR filter). Bistability and hysteresis require a nonlinear unit, and any saturating recurrent unit with loop gain above one produces both.
10. *Non-anthropomorphic physical attention?* Gibbs weighting of competing modes with saturation (F12).

The investigator's closing question, whether attention is needed for the dynamics or only for observing them, has a precise form: it is the question $C_E \equiv 0$ or $C_E \neq 0$, which no observational statistic settles (Proposition 5) and which Protocol B and interventions on the memory substrate can test.

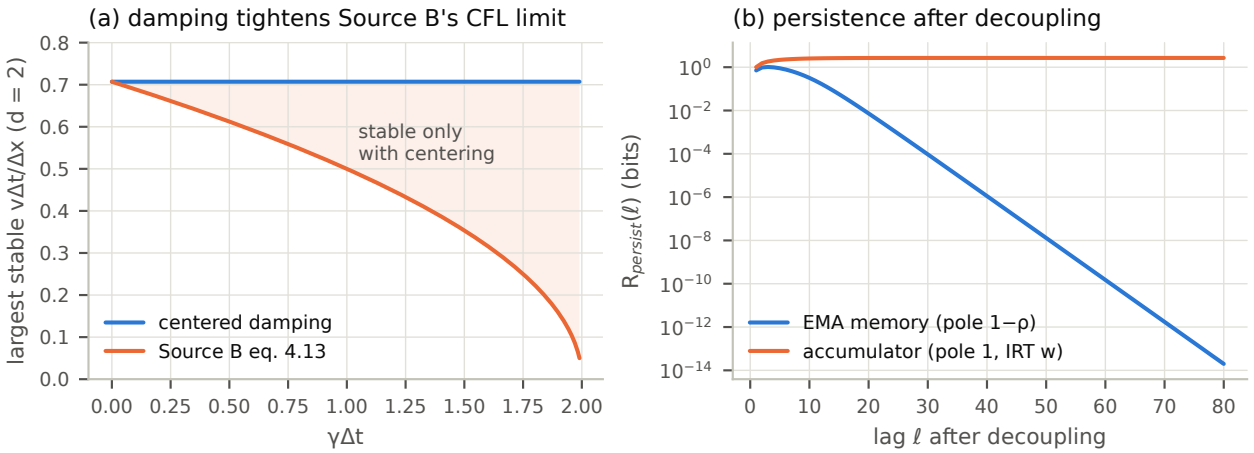


Figure 5: (a) Largest stable Courant number in 2D for centered damping and for Source B's eq. (4.13). (b) $R_{\text{persist}}(\ell)$ after decoupling for an EMA memory and an accumulator; under H1 it is identically zero (not drawable on a log scale).

9 Engineering: the kgirl integration

The package `src/kgirl/iwme` follows the conventions of `kgirl.nihiline`: numpy-only modules with optional sympy, a command-line interface, unittest suites, and figures generated from code.

Module	Content
`attention`	P2, P2′, Lemma 8, P3, P3′, hysteresis, gated forgetting
`cascade`	binomial, Markov and memory cascades; transfer operator; Hutchinson fixed point
`irt`	IRT metric, determinant, rank, singular circles, curvature (three routes), Source C transcriptions
`wake`	centered and Source B schemes, CFL and order, energy identity, zero-order hold, Source B checks
`sources`	Source D Tier 1–2 checks, coherence measures
`core`	repaired core (2), exact R_{persist}
`verify`	the 47-check registry; table, JSON and LaTeX output (Appendix A)
`plots`	the five figures of this report

The harness MCP server gains three tools: `iwme_verify`, `iwme_irt_point` and `iwme_attention`. Before editing `harness/mcp_server.py`, its blast radius was computed with the repository's Atlas index: five dependents, all inside kgirl (two test modules, the CLI and its entry point, the nihiline tests); the harness, nihiline and evolve suites pass after the change. The 70 IWME unit tests pass in about 15 s. Branch `claude/iwme-phase3` holds the work; merging is left to the investigator.

10 Checkpoint and next actions

Q1 Definition of "intra-dimensional" and "emission": open (candidate readings in F6 and Section 6.1).

Q2 Sources B, C, D: supplied and audited; Source E received and outside scope.

Q3 Narrowing to the inductive-bias claim with $C_E \neq 0$ as extension M4: clarification received (Section 6.2); a formal decision is still requested.

Q4 Execution environment: used; every number in this report is computed.

Q5 Physical data or apparatus for Experiments B, C, F: open. Phase 4 proceeds simulation-first.

Q6 Authorship and venue: open.

Phase 4. (A) multifractal experiment with pre-registered symbolization and parameter-count-matched model comparison; (B) Protocol B on a nonlinear simulated plant, then on a physical pattern-forming system if Q5 allows; (E) attention ablation in a mode-competition system (Swift–Hohenberg with Gibbs mode weighting, F12), sweeping T through T_c and measuring hysteresis at extrapolated sweep rate; (F) wake perturbation with the zero-order-hold source and pre-registered P_E.

11 Adversarial reviewer's verdict

Draft 1.0 remains unready for submission: its core, as written, conflates observation with actuation and leaves the loop open, and its presence criterion measures coupling. The repaired core of Section 6.1 is coherent, and the attention mechanism now carries precise, falsifiable predictions: a pitchfork at T_c, algebraic slowing with exponent $\frac{1}{2}$, hysteresis area scaling as $(1-T)^2$, and flip or Neimark–Sacker crossings set by the key–value geometry. The supplied sources must enter with care. Source C's IRT is a well-defined manifold whose metric is nondegenerate off explicitly located circles (the two folds at $\psi = 0$ and two circles at each zero of κ; ten for the damped kernel on $0 \leq \psi \leq 6$), and its published metric, Christoffel symbols, curvature and shadow integral are wrong. Source B's categorical statements are ill-typed and its wake system has six identified defects; its finite fixed-point content survives as the Hutchinson model. Source D supplies no usable measurement. The strongest recoverable material is the H1/H2 axis, the attention bifurcation, the identifiability statement, and an exact persistence criterion that tells the program where evidence can come from.

Evidence and speculation. Established (L0): every proposition, lemma and numerical check above. Extrapolation (L1): that the gated and memory-cascade variants describe classes of physical systems. Hypothesis (L2): the parametric IWME claim of Section 6.2 and M4. Speculative (L3): presence as more than state incompleteness; any reading of the IRT as a communication device.

A Verification registry

Generated from the repository root by

```
PYTHONPATH=src python -m kgirl.iwme verify --latex research/iwme/verification_table.tex
```

"Observed" is the program's output. A refutation of a source claim appears as a passing check. All 47 checks pass.

Id	L	Claim	Observed	Result
V01	0	P2: with the attention sequence fixed, ∂H_t/∂H_{t−k} = (1−ρ)^k	0.000976562	PASS

Id	L	Claim	Observed	Result
V02	0	P2′: closed loop (scores read H), ∂H_t/∂H_0 is the product of $(1-\rho)$I + (ρ/T)Cov_a(v,k)	0.364759 (FD 0.364759), ratio to open loop 374	PASS
V03	0	Closed-loop retention time at H=0 diverges at $T\to1^+$ and tends to τ_mem as $T\to\infty$	T=1.05: 41.498, T=1.5: 5.485, T=3: 2.466, T=1e+06: 1.443	PASS
V04	0	Lemma: sup_a $\|$Cov_a(v,k)$\| \le$ D_v D_k / 4 (contraction for T > D_v D_k / 4)	max ratio 0.9581	PASS
V05	0	The D_v D_k/4 bound is attained (two-point example at H=0), so T_contract = T_c = 1 there	$\|$Cov$\|$ = 1.000000000000, bound 1.000000000000	PASS
V06	0	P3 at T=0.5, ρ=0.5: fixed points 0 (unstable) and $\pm$H* (stable)	H* = 0.957504; μ = 0.5832, 1.5000	PASS
V07	0	Near T_c: H* $\approx$ T$\sqrt{}$(3(1−T)) (mean-field exponent $\frac{1}{2}$)	T=0.99: 1.006054, T=0.999: 1.000601, T=0.9999: 1.000060	PASS
V08	0	Critical slowing at T=1: H_t $\sqrt{}$t $\to \sqrt{}(3/(2\rho))$ (algebraic decay)	1.731982 at t = 1e5	PASS
V09	0	Log–log decay slope: $-\frac{1}{2}$ at T=1; 0 for T<1 (lock-in); steep for T>1	T=0.8: 0.000, T=1.0: -0.490, T=1.2: -7.491	PASS
V10	0	Static (adiabatic) hysteresis area > 0 iff T < 1	T=0.6: 0.658, T=0.8: 0.173, T=0.95: 0.0111, T=1.01: 1.39e-20, T=1.3: -4.74e-18	PASS
V11	0	Hysteresis area $\propto (1-T)^2$ near T_c (mean field)	1.975	PASS
V12	0	Anti-aligned values: flip (period-2) bifurcation at T_f = $\rho/(2-\rho)$	0.1265 / 3.52e-16	PASS
V13	0	P3′: crossing type set by the spectrum of C_0 = Cov_uniform(v,k)	aligned: pitchfork at 0.5; anti: flip at 0.1667; rotated: Neimark–Sacker at 0.2887	PASS
V14	1	Gated forgetting makes open-loop retention attention-dependent	0.599 vs 4.31e-06	PASS
V15	0	Report table: D_q at p = 0.7 (hand-computed values, 4 d.p.)	-1: 1.1258, 0: 1.0000, 0.5: 0.9385, 1: 0.8813, 2: 0.7859, 3: 0.7172, inf: 0.5146, -inf: 1.7370	PASS
V16	0	Box counting on the exact level-16 measure reproduces the closed form	max 2.2e-16	PASS
V17	0	H_n = n h2(p) exactly (no limit needed)	max 1.8e-15	PASS
V18	0	Markov cascade: Z_n = $\pi_0{}^{\mathsf{T}}$M_q^{n}1 (enumeration) and i.i.d. limit = binomial	Z_n rel err 7.0e-16; i.i.d. $\tau(2)$ err 0.0e+00; D_2 = 0.8279	PASS
V19	1	EMA-memory cascade: transfer operator gives τ(q); memory moves D_2 off the matched memoryless value	$\tau(2)$ transfer 0.49709, enumeration (n 16→22) 0.49815; D_2 memoryless 0.6940 vs 0.4971	PASS
V20	0	EFL fixed point model: Hutchinson operator contracts W1 by exactly $\frac{1}{2}$; binomial is its fixed point	max abs(ratio − $\frac{1}{2}$) 2.3e-14; fixed-point err 1.7e-18	PASS
V21	0	Source B: D_eff = limsup log N(r)/log λ^r with N(r) ~ λ^r is identically 1	1.000000000000 … 1.000000000000	PASS
V22	0	Closed-form induced metric = $J^{\mathsf{T}}J$ (numeric, 2 kernels × 5 points)	1.8e-15	PASS
V23	0	Symbolic proof: $J^{\mathsf{T}}J$ = closed form and det g = u^{2}e^{−2$\alpha\psi$}[r^{2}e^{−2$\alpha\psi$}D^2 + κ^2g_$\varphi\varphi$]	g_thetatheta: True, g_thetaphi: True, g_thetapsi: True, g_phiphi: True, g_phipsi: True, g_psipsi: True, det: True	PASS
V24	0	Source C A.1 is wrong in g_$\varphi\varphi$, g_$\psi\psi$ and g_$\varphi\psi$ (g_$\theta\theta$ and g_$\theta\varphi$ = g_$\theta\psi$ = 0 are right)	theta-theta: 0, phi-phi: 0.0358, psi-psi: 0.0804, phi-psi true value: -0.563	PASS
V25	0	Jacobian rank 3 off the predicted circles; rank 2 and det g = 0 exactly on them	10 circles (2 at ψ=0, 2 per zero of κ: ψ_k = [0.7854, 2.3562, 3.927, 5.4978])	PASS
V26	0	Scalar curvature: Gauss equation = intrinsic Riemann = warped product 2K_Σ − 2Δ_$\Sigma\rho/\rho$	3.2e-06	PASS
V27	0	Curvature depends on κ only through κ and κ': changing $\kappa''(\psi_0)$ leaves Scal(ψ_0) unchanged	$\kappa'' \in \{0, 1, -4, 14\}$: spread 1.6e-08	PASS

Id	L	Claim	Observed	Result
V28	0	Large constant κ: Scal $\to$ 2K of the ψ-slice torus, which changes sign at cos φ = 0 (Gauss–Bonnet)	κ_0=10: 1.6e-01, κ_0=100: 1.6e-03, κ_0=1000: 1.6e-05; sign rule True	PASS
V29	0	Source C: 'negative curvature' is false; Scal takes both signs over the chart	constant κ=1: 67% neg / 33% pos; e^{-0.2ψ} cos 2ψ: 56% neg / 44% pos	PASS
V30	0	Source C eq. 29 disagrees with the curvature in value and sign	sign differs at 2/5 points; e.g. true +2.736 vs eq. 29 -8.186	PASS
V31	0	Source C eq. 28: Γ^ψ_$\theta\theta$ has the wrong sign even for its own (diagonal) metric	Γ^ψ_$\theta\theta$ printed -1.2778 vs own +1.2778; Γ^ψ_$\varphi\varphi$ +0.3579 vs +0.0333	PASS
V32	0	Source C eq. 16 is not an antiderivative of eq. 15 (and w(0) $\neq$ 0)	exact residual 1.0e-10; eq. 16 residual 2.921; eq. 16 w(0) = -5.0	PASS
V33	0	CFL: centered limit 1/$\sqrt{}$d for every $\gamma\Delta$t; Source B's scheme $\sqrt{}$((1$-\gamma\Delta$t/2)/d)	max abs diff 5.8e-11	PASS
V34	0	Temporal order: centered 2, Source B (4.13) 1	1.998 and 1.005	PASS
V35	0	Discrete energy identity of the centered scheme (2D, damped, forced)	residual 9.3e-15	PASS
V36	0	Source B Prop. 4.2 is false: curl $\nabla\theta$ carries 2πm; the flux of $\nabla\theta$ ($\int\Delta\theta$) is 0	circulation/2π = [1.0, 1.0, 1.0, 1.0, 1.0, 1.0]; max abs flux 0.0e+00	PASS
V37	0	Source B (4.2c) does not preserve c $\in$ [0,1]	c∞ = -0.5, crosses 0 at t = 1.0986 (= ln 3)	PASS
V38	0	Source B Prop. 4.1: energy need not decay with stationary S, c (static source, start at rest)	E: 5.62e-08 $\to$ max 3.57e-05	PASS
V39	0	No localized steady wake without a restoring term; (4.9) holds only with mass γ_m W	W(0) = [0.5, 1.0, 2.0]; (4.9) max err 7.3e-05	PASS
V40	0	Decoherence pulse: uniform-mode displacement β/γ, independent of σ (Source B: $\beta\sigma/\gamma$)	σ=2: 2.000000; σ=0.5: 2.000000	PASS
V41	0	Lagrangian (4.4) does not generate (4.2a)	EL $-$ (4.2a) = ($\alpha - \chi$v^2)S_xx + v$^2\omega^2$W $-$ v$^2\xi$(1 $-$ c) $-\beta\partial$_t c (restoring term, no ∂_t c source)	PASS
V42	0	2D point source: $\int\nabla$W·∇W diverges as ln(1/h)/(2π), outside Theorem 4.1's H^1	0.7329 $\to$ 1.4659	PASS
V43	0	Source D Tier 1: reported Walsh coefficients belong to a different input; Parseval passes for both	WHT[1.0, 1.0, 1.0, -1.0, 1.0, 1.0, 1.0, 1.0] = [6.0, 2.0, 2.0, -2.0, -2.0, 2.0, 2.0, -2.0]; reported = transform of [1.0, 1.0, 1.0, -1.0, -1.0, 1.0, 1.0, 1.0]	PASS
V44	0	Source D Tier 2: φ(a^k) = φ(a)^k for φ = mod 3 on Z9 cannot fail	0 failures in 279; unital Z4$\to$Z3 map exists: False	PASS
V45	0	Source D Prediction 5 metric is maximal for a constant signal	constant 1.000; chaotic noise 0.011	PASS
V46	0	Source B eq. 4.10 inverts agreement; spectral share is sign-blind; the resultant separates the antipodal pair	identical: 7.4e-17/1/1; orthogonal: 1/0/0; antipodal_pair: nan/1/0; two_clusters: 0.33/0.33/0.33	PASS
V47	0	R_persist: 0 under H1; > 0 with actuation; 0 again given (X,H); EMA decays, accumulator saturates	H1 0.0e+00; X 1.366 bits; X,H 1.0e-14; lag 60: EMA 1.5e-10, accumulator 2.68 bits	PASS

B Notation

X_t physical state; H_t memory; A_t attention weights; Y_t interaction signal; E_t environment parameters; U_t controls; C_E actuation map; $\Pi_{E,\lambda}$ perspective map; T attention temperature; ρ write rate; γ wake damping (Source B's λ); λ observational scale; H$[\cdot]$ entropy. IRT: $u = R - r\cos\varphi$,

$\rho = ue^{-\alpha\psi}$ (radius of the θ-circle; the memory write rate ρ never appears in the IRT sections), $D = \beta r \sin^2\varphi \cosh\beta\psi + \alpha u \cos\varphi \sinh\beta\psi$.

References

[1] K1ll. *Intra-dimensional Working Memory Emission*, Draft 1.0. Unpublished manuscript.

[2] Anonymous. *The Emergent Fractal Law and the 3^{627} Framework: A Categorical Theory of Self-Similar Information Geometry*. Unpublished manuscript, 21 October 2025.

[3] Neurotronic Phase Caster Research Collective. *Geometry as Emergent Uni-telligence: The Informational Handshake of Higher-Dimensional Coherence Measurement*. Unpublished manuscript, 7 November 2025.

[4] J. Edwards (with D. J. McConnell). *The Convergence Point: A Developmental Framework for Machine Consciousness*. Manuscript.

[5] M. Oo et al. *From Triangle to Form: Bridging Discrete Relational Networks to Continuum Spacetime and Hadronic Droplets via E8 Holographic Hydrodynamics*. Manuscript, 2026.

[6] C. E. Shannon. A mathematical theory of communication. *Bell Syst. Tech. J.* 27:379–423, 623–656, 1948.

[7] R. C. Conant, W. R. Ashby. Every good regulator of a system must be a model of that system. *Int. J. Syst. Sci.* 1(2):89–97, 1970.

[8] B. B. Mandelbrot. Intermittent turbulence in self-similar cascades. *J. Fluid Mech.* 62:331–358, 1974.

[9] H. G. E. Hentschel, I. Procaccia. The infinite number of generalized dimensions of fractals and strange attractors. *Physica D* 8:435–444, 1983.

[10] T. C. Halsey, M. H. Jensen, L. P. Kadanoff, I. Procaccia, B. I. Shraiman. Fractal measures and their singularities. *Phys. Rev. A* 33:1141–1151, 1986.

[11] A. Rényi. On measures of entropy and information. *Proc. 4th Berkeley Symp.* 1:547–561, 1961.

[12] D. Ruelle. *Thermodynamic Formalism*. Addison-Wesley, 1978.

[13] D. Lind, B. Marcus. *An Introduction to Symbolic Dynamics and Coding*. Cambridge University Press, 1995.

[14] J. E. Hutchinson. Fractals and self-similarity. *Indiana Univ. Math. J.* 30:713–747, 1981.

[15] F. Takens. Detecting strange attractors in turbulence. *Lecture Notes in Math.* 898:366–381, 1981.

[16] R. E. Kalman. A new approach to linear filtering and prediction problems. *J. Basic Eng.* 82:35–45, 1960.

[17] C. W. J. Granger. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* 37:424–438, 1969.

[18] T. Schreiber. Measuring information transfer. *Phys. Rev. Lett.* 85:461–464, 2000.

[19] J. Pearl. *Causality*, 2nd ed. Cambridge University Press, 2009.

[20] J. Rissanen. Modeling by shortest data description. *Automatica* 14:465–471, 1978.

[21] D. Bahdanau, K. Cho, Y. Bengio. Neural machine translation by jointly learning to align and translate. *ICLR*, 2015.

[22] A. Vaswani et al. Attention is all you need. *NeurIPS* 30, 2017.

[23] F. A. Gers, J. Schmidhuber, F. Cummins. Learning to forget: continual prediction with LSTM. *Neural Comput.* 12:2451–2471, 2000.

[24] J. J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *PNAS* 79:2554–2558, 1982.

[25] Yu. A. Kuznetsov. *Elements of Applied Bifurcation Theory*, 3rd ed. Springer, 2004.

[26] Y. Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, 1984.

[27] A. M. Turing. The chemical basis of morphogenesis. *Phil. Trans. R. Soc. B* 237:37–72, 1952.

[28] M. C. Cross, P. C. Hohenberg. Pattern formation outside of equilibrium. *Rev. Mod. Phys.* 65:851–1112, 1993.

[29] N. D. Mermin. The topological theory of defects in ordered media. *Rev. Mod. Phys.* 51:591–648, 1979.

[30] J. C. Strikwerda. *Finite Difference Schemes and Partial Differential Equations*, 2nd ed. SIAM, 2004.

[31] B. O'Neill. *Semi-Riemannian Geometry with Applications to Relativity*. Academic Press, 1983.

[32] M. P. do Carmo. *Riemannian Geometry*. Birkhäuser, 1992.

[33] P. Deligne, P. Griffiths, J. Morgan, D. Sullivan. Real homotopy theory of Kähler manifolds. *Invent. Math.* 29:245–274, 1975.

[34] J. L. Walsh. A closed set of normal orthogonal functions. *Amer. J. Math.* 45:5–24, 1923.

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